

Process Control



Group 1

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Introduction

- **Control the water level using the pump and outlets**
 - **Level control is used in industry to maintain level for liquid-liquid interfaces and liquid-vapour interfaces.**
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Theory

- To keep the level constant, the flow rate in, must equal the flow rate out.
- When draining the tank, there may still be water left, but we assume it is empty.
- The experiment varies manual and automatic control of the pump speed to keep the level constant 

Apparatus

- Pump, upper and lower tank. Solenoid valve and manual valve.

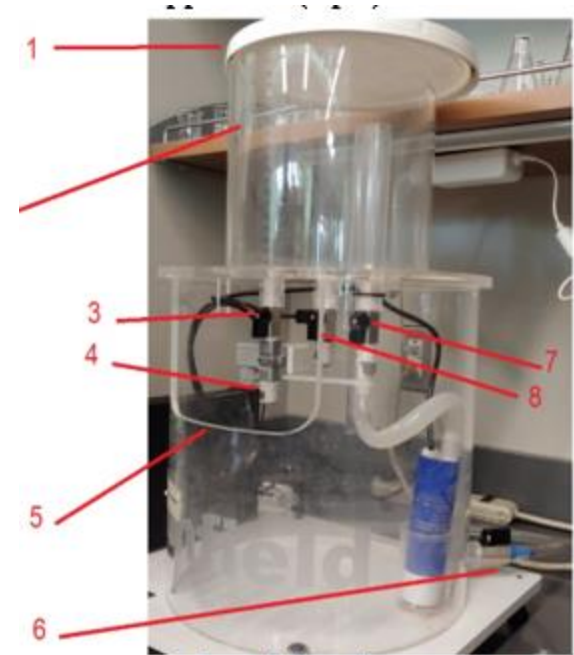


Figure 1: Sump Pump apparatus.

Material & Supplies

- PCT50 pump 
- Water is used as the fluid
- Computer to control pump speed and valves

Procedure

- Experiment A
 - Use manual control to maintain a constant level
 - Observe disturbances (opening valves)
- Experiment C 
 - Use P only controller
 - Use P+I controller

Experimental Challenges



- Remove air bubbles from the pump before starting lab.
- Check if the level sensor is reading the level correctly throughout the lab.
- For Exercise C, P only, the $P=200\%$ will always cause errors. This is the result that is supposed to happen

Safety

- Emergency overflow drain if water is too high
- Be careful of water near electrical equipment

Comment Summary

Page 1

1. 5/5
2. 91/100

Page 2

3. That's not the objective
4. 12/15

Page 3

5. What about P and I controllers?
6. 13/15

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7. PCT controller? Computer?
8. 13/15

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9. Part of apparatus
10. 14/15

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11. Few more steps need to be shown
12. 14/15

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13. 10/10

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14. 10/10